Davis Clark

404-823-8585 | daviskclark@gmail.com | https://www.linkedin.com/in/daviskclark/ | https://github.com/davisclrk

Education

Georgia Institute of Technology — Atlanta, GA

Master of Science in Computer Science, GPA: 4.0/4.0 — *Expected May 2027*

Bachelor of Science in Computer Science, GPA: 3.96/4.0 — *May 2026*

- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Object-Oriented Design, Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Architecture, Databases, Networking
- Graduate TA, Machine Learning: built homeworks & autograders from scratch for new section, co-designed exams

Experience

Google — May 2026 – August 2026

Software Engineering Intern — *New York, NY*

- Designed **6 stage, zero downtime** migration of live encryption key table from legacy database to Cloud Spanner
- Implemented dual writes and shadow reads to cut over safely, with automatic fallbacks on every read & write path
- Built dynamic read routing between async mirror & primary to **bound stale reads**, with failover on mirror errors
- Validated data parity across legacy, primary, and mirror DBs, logging mismatches after accounting for replica lag

DoorDash — May 2025 – August 2025

Software Engineering Intern — *San Francisco, CA*

- Built Apache Iceberg backfill migration service with transformation logic to normalize inconsistent legacy schemas
- Engineered multithreaded PySpark backfill pipeline with custom Java listeners to log real-time backfill progress
- Cut backfill cost by **76%** and time by **57%** by optimizing writes and batching updates to reduce lock acquisitions
- Reduced manual intervention by **96%** by streamlining backfills with Java REST APIs and data validation layers
- Improved reliability of Flink routing system handling **130k+** telemetry req/s with robust exception handling logic

Hubbell Incorporated — May 2024 – August 2024

Software Engineering Intern — *Greenville, SC*

- Reduced server load by **68%** by replacing polling w/ WebSocket pub-sub notifications when downloads complete
- Cut quote form retrieval time by **92%** by building C# REST APIs and React dashboard to automate downloads
- Built C# microservice and Azure CRON job to vet & send OTPs to **18,000**+ legacy accounts, saving **30+** hrs/qtr

Projects

Tourette's Detection Wristband | Python, Flutter, Dart, C++, Firebase

- Created wristband to detect Tourette's Syndrome tic attacks in real time from motion and biometric sensor data
- Trained LOF anomaly detector and LSTM audio classifier in TensorFlow, achieving **92%** accuracy on speech data
- Published **co-first authored research paper** on models & applications in Journal of Medical Internet Research
- Developed mobile companion app for patient dashboard and caretaker alert notifications with Flutter and Firebase

Agentic Chinese Tutor | React Native, TypeScript, Python, PyTorch

- Created interactive LiveKit voice tutor to teach Chinese w/ LLM controlling tutoring sessions and learner memory
- Achieved **97% accuracy** on tone and pronunciation analysis by training supervised **HuBERT** classifier heads
- Developed real-time ASR, TTS, and syllable alignment pipeline with parallel scoring to minimize per-turn latency

Distributed Semantic Search Database | Go, gRPC, Protobuf, SQLite

- Built sharded, replicated vector database for semantic article search with parallel top-k query fanout over gRPC
- Implemented HNSW nearest neighbor index with **99.6% recall@10**, **200x** faster than brute force on 100K vectors
- Added synchronous replication with rollback on replica failure, read failover to replicas, and SQLite persistence

Community Service Hub | React, JavaScript, Django, Python, PostgreSQL

- Developed fullstack CRUD webapp to aggregate local volunteering events and track service hours for volunteers
- Onboarded **750+ students** to digitally record volunteering hours by integrating platform with local high school

Skills

Languages: C/C++, Python, Java, JavaScript, C#, Go, Dart, HTML/CSS, SQL
Technologies: React, Django, Node.js, .NET, Flutter, Docker, PyTorch, Spark, Flink, gRPC, GCP, Azure
Concepts: Web Frameworks, Software Engineering, Frontend, Backend, Fullstack, Machine Learning, Human Computer Interaction, Data Science, Data Analysis, Distributed Systems, Cloud Computing, Virtual Memory, Multithreading, Parallel Programming, Microservice Architecture, Agile, DevOps, CI/CD, Blockchain, Statistics, Probability, NLP